

DERECK MÉNDEZ

DATA ANALYST

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PROFESSIONAL SUMMARY

Data Analyst with experience building data pipelines, predictive models, and automated retraining systems for large-scale contact center operations. I combine software engineering practices (version control, containers, best practices) with statistical analysis, machine learning, and data visualization (Python, SQL, Power BI) to turn operational data into business decisions, and I am expanding my experience into production model deployment and monitoring (Docker, Kubernetes, CI/CD).

EXPERIENCE

Engineering Consultant – ACF Technologies (Jun 2026 – Present)

- Led the full refactoring of the demand and service-time forecasting pipeline (Python, SQL Server) and the agent scheduling module that consumes its predictions: reorganized it into modules, documented the business logic, and fixed legacy bugs without disrupting a live production system.
- Optimized the forecasting model's automated retraining process by introducing champion/challenger evaluation to promote only better-performing models, achieving incremental accuracy gains over a model that was already optimized.
- Designed and built, from scratch, an end-to-end system to predict demand, wait times, and service times for new clients: database modeling, APIs, automated testing, and containerized deployment (Docker, Kubernetes).
- Introduced software engineering practices previously absent from the team's projects: code and data version control, continuous integration and deployment (CI/CD), and the use of AI tools to accelerate development.

Training and Development Coordinator — Almacenes Vidrí S.A. de C.V. (Jul 2023 – Jun 2026)

- Automated the extraction, transformation, and consolidation of training data from multiple APIs and operational reports using Python, eliminating manual processes and significantly reducing processing time.
- Designed relational data models and Power BI dashboards to track strategic KPIs for compliance, participation, and performance, improving visibility for decision-making.
- Led the creation of an internal training school and analyzed training and performance metrics to optimize program effectiveness.

FEATURED PROJECTS

Data Science Salary Analysis (End-to-End)

- Built a Python ETL pipeline to extract, clean, and consolidate global Data Science salary data, and performed an EDA to identify compensation patterns by experience, location, and work modality.
- Designed SQL queries to analyze salary trends and gaps between roles, and developed an interactive Power BI dashboard to communicate insights at a strategic level.

Customer Churn Prediction with MLOps Pipeline

- Built an end-to-end customer churn prediction system: compared XGBoost, LightGBM, and CatBoost during model selection, and used MLflow for model tracking/registry, DVC for data versioning, and an automated drift detection and retraining pipeline.
- Deployed an inference API (validated with Pydantic) and a monitoring dashboard in Streamlit (Hugging Face Spaces), with the rest of the pipeline orchestrated locally via Docker Compose.

Data Engineering Pipeline for E-commerce (In progress)

- Developing a data pipeline with Polars and Airflow orchestration for the e-commerce/retail domain, applying data engineering principles to a real business case.

EDUCATION

Software Engineering and Data Science (In progress) — Universidad Evangélica de El Salvador

Python Programming Specialization (2024) — ISEADE-FEPADE

TECHNICAL SKILLS

Languages and tools: Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL, Git, Docker, Kubernetes

Machine Learning and MLOps: Predictive models, XGBoost, LightGBM, CatBoost, MLflow, DVC, CI/CD, REST APIs, drift detection

Visualization and BI: Power BI (DAX, Power Query), advanced Excel, executive dashboards

Data analysis: EDA, ETL, data transformation and modeling